LAB-8

Title- Application of K-Means Clustering Algorithm for Determining the Optimal Number of Clusters

Objectives

1. To understand the concept of **K-Means clustering**.
2. To implement the K-Means clustering algorithm on a suitable **open-access dataset**.
3. To understand the concept of **centroids and cluster formation**.
4. To determine the **optimal number of clusters** using the **Elbow Method or Silhouette Method**.
5. To visualize and analyze the clustering results.
6. To evaluate the quality of the obtained clusters using appropriate clustering metrics.

Problem Statement- Apply the K-Means clustering algorithm on a suitable open-access dataset to obtain the optimal number of clusters using the Elbow or Silhouette method.

Theory

K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to divide a dataset into a predefined number of groups called clusters. The main objective of K-Means is to group similar data points into the same cluster and keep dissimilar data points in different clusters. Each cluster is represented by a centroid, which is the mean position of all the data points belonging to that cluster. The value of K represents the number of clusters to be formed.

Working of K-Means

K-Means generally works through the following steps:

1. Select the number of clusters K.
2. Initialize K cluster centroids.
3. Assign each data point to the nearest centroid.
4. Calculate new centroids based on the assigned data points.
5. Reassign the data points to the nearest new centroid.
6. Repeat the process until the centroids become stable or the algorithm converges.

Objective of K-Means

K-Means tries to minimize the distance between data points and their corresponding cluster centroid. The commonly used objective is **Within-Cluster Sum of Squares (WCSS)**, also called **inertia**. A lower WCSS indicates that the data points are closer to their respective cluster centroids.

Determining the Optimal Number of Clusters

Selecting the appropriate value of **K** is important because choosing too few or too many clusters may produce poor clustering results.

Two commonly used methods are:

1. Elbow Method

The Elbow Method calculates the WCSS for different values of K, such as K = 2, 3, 4, 5, etc. The WCSS generally decreases as K increases. The optimal value of K is selected at the point where the decrease in WCSS becomes significantly smaller, creating an "elbow" in the graph.

2. Silhouette Method

The Silhouette Method measures how well each data point fits within its assigned cluster. The Silhouette Score generally ranges from -1 to +1.

- **Close to +1:** Data points are well matched to their own cluster.
- **Around 0:** Clusters may overlap.
- **Less than 0:** Data points may be assigned to the wrong cluster.

Conclusion

The K-Means clustering algorithm is successfully applied to a suitable open-access dataset. Different values of K are evaluated using the Elbow Method or Silhouette Method to determine the optimal number of clusters. The selected K is then used to perform final clustering, and the resulting clusters and centroids are visualized and analyzed.

Viva Questions

1. What is K-Means clustering?
2. Why is K-Means called an unsupervised learning algorithm?
3. What does **K** represent in K-Means?
4. What is a cluster?
5. What is a centroid?
6. How does the K-Means algorithm work?
7. What is the objective of K-Means clustering?
8. What is WCSS?
9. What is the Elbow Method?
10. Why is the Elbow Method used?
11. What is the Silhouette Score?
12. What is the range of the Silhouette Score?
13. Which Silhouette Score indicates better clustering?
14. How do you select the optimal value of K?
15. What are the advantages of K-Means clustering?
16. What are the limitations of K-Means?